8 Factor Analysis: Practical Methods

MA2003B Application of Multivariate Methods in Data Science

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where:

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where:

- X is a random vector of p observed variables (indicators).
- μ is a constant vector of dimension p.
- L is a $p \times m$ constant matrix known as the matrix of *factor loadings.*
- $F \sim \mathcal{N}(0, I_m)$ is a vector of m latent factors (unobserved variables) that are assumed to be normally distributed with mean zero and identity covariance matrix.
- $\varepsilon \sim \mathcal{N}(0, \Psi)$ is a vector of unique errors (specific variances) that are also assumed to be normally distributed with mean zero and diagonal covariance matrix $\Psi = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_p^2)$.

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Notice that, since $\varepsilon \sim \mathcal{N}(0, \Psi)$, this model is completely specified by the parameters μ, L, and Ψ. In this chapter we will discuss practical methods for estimating these parameters from data, as well as methods for estimating the latent factors F (known as *factor scores*) for our observations once the model parameters have been estimated.

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Assessing Factorability

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Several statistical tests and measures can be used to evaluate the factorability of a dataset:

1. **Correlation Matrix Examination:** A preliminary step is to visually inspect the correlation matrix of the observed variables. Look for patterns of high correlations (e.g., > 0.3) among variables, which suggest that they may share common underlying factors.

2. **Bartlett's Test of Sphericity:** This test evaluates the hypothesis that the correlation matrix is an identity matrix, which would indicate that the variables are uncorrelated and unsuitable for factor analysis. A significant result (p-value < 0.05) suggests that the variables are correlated enough to proceed with factor analysis.

3. **Kaiser-Meyer-Olkin (KMO) Measure of Sampling Adequacy:** The KMO statistic measures the proportion of variance in the observed variables that can be attributed to underlying common factors, relative to variance that is unique to each variable. It ranges from 0 to 1, with values closer to 1 indicating that factor analysis is likely to be useful. A KMO value above 0.6 is generally considered acceptable for factor analysis.

Deciding the Number of Factors

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Several methods can be employed to decide on the number of factors:

1. **Eigenvalue Criterion (Kaiser Criterion):** Retain factors with singular values greater than 1. This method is based on the idea that a factor should explain at least as much variance as a single observed variable.
2. **Scree Plot:** A scree plot displays the singular values in descending order. Look for an "elbow" point where the slope of the curve changes dramatically. The number of factors to retain is typically determined by the number of points before this elbow.
3. **Parallel Analysis:** This method involves comparing the observed singular values with those obtained from random data of the same size. Factors are retained if their singular values exceed those from the random data.
4. **Model Fit Indices:** For more advanced factor analysis methods (e.g., confirmatory factor analysis), various fit indices (e.g., RMSEA, TLI) can be used to assess how well the model fits the data for different numbers of factors.

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1. **Principal Axis Factoring (PAF):** This method focuses on the shared variance among variables and is suitable when the goal is to identify underlying constructs. It iteratively estimates communalities and extracts factors based on the reduced correlation matrix.
2. **Maximum Likelihood (ML) Factor Analysis:** This method estimates factor loadings by maximizing the likelihood of the observed data given the model. It allows for statistical testing of hypotheses about the factor structure and provides goodness-of-fit indices.

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Common rotation methods include:

1. **Varimax Rotation:** An orthogonal rotation method that maximizes the variance of squared loadings within each factor, leading to a simpler structure where each variable loads highly on one factor and low on others.
2. **Quartimax Rotation:** Another orthogonal rotation method that simplifies the rows of the loading matrix, aiming to have each variable load highly on as few factors as possible.
3. **Equamax Rotation:** A hybrid orthogonal rotation method that combines the goals of varimax and quartimax, balancing the simplification of both rows and columns of the loading matrix.
4. **Promax Rotation:** An oblique rotation method that allows factors to correlate. It starts with a varimax rotation and then raises the loadings to a power to achieve a simpler structure.

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Several methods can be used to compute factor scores:

1. **Regression Method:** This method estimates factor scores by regressing the observed variables on the estimated factor loadings. It provides unbiased estimates of factor scores but may not preserve the original variance of the factors.
2. **Bartlett Method:** This method computes factor scores by minimizing the residual variance of the observed variables. It tends to produce factor scores that are more reliable and have lower measurement error.
3. **Anderson-Rubin Method:** This method generates factor scores that are uncorrelated and standardized. It is particularly useful when the factors are assumed to be orthogonal.

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Several statistical tests and fit indices can be used to evaluate model fit:

1. **Chi-Square Test of Model Fit:** This test evaluates the null hypothesis that the model-implied covariance matrix matches the observed covariance matrix. A non-significant result (p-value > 0.05) suggests that the model fits the data well.
2. **Root Mean Square Error of Approximation (RMSEA):** This index measures the discrepancy between the observed and model-implied covariance matrices per degree of freedom. Values less than 0.06 indicate a good fit.
3. **Tucker-Lewis Index (TLI):** This index compares the fit of the specified model to a null model. Values greater than 0.95 indicate a good fit.
4. **Standardized Root Mean Square Residual (SRMR):** This index measures the average discrepancy between observed and predicted correlations. Values less than 0.08 indicate a good fit.

Visualizing the Model

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Common visualization techniques include:

1. **Scree Plot:** A scree plot displays the singular values in descending order, helping to visualize the number of factors to retain.
2. **Factor Loading Plot:** A scatter plot of factor loadings can help visualize how variables load on different factors, especially after rotation.
3. **Biplot:** A biplot combines factor scores and loadings in a single plot, allowing for the visualization of both observations and variable relationships in the factor space.
4. **Heatmap of Loadings:** A heatmap can be used to visualize the magnitude of factor loadings across variables and factors, highlighting patterns and clusters.

Remark

Notice that Factor Loading Plots and Biplots only make sense when we have 2 or 3 factors, otherwise we cannot visualize them properly.

Example: Factor Analysis of the Holzinger-Swineford Dataset

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This dataset contains scores from a battery of 9 cognitive tests administered to 301 students in grades 7 and 8.

The tests were designed to measure 3 latent abilities: visual/spatial, verbal, and speed. For this example we will focus only on tests x4 to x9 so that we can fit a two-factor model and visualize the results in a biplot.

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The tests were designed to measure 3 latent abilities: visual/spatial, verbal, and speed. For this example we will focus only on tests x4 to x9 so that we can fit a two-factor model and visualize the results in a biplot.

The exact description of each test is given below:

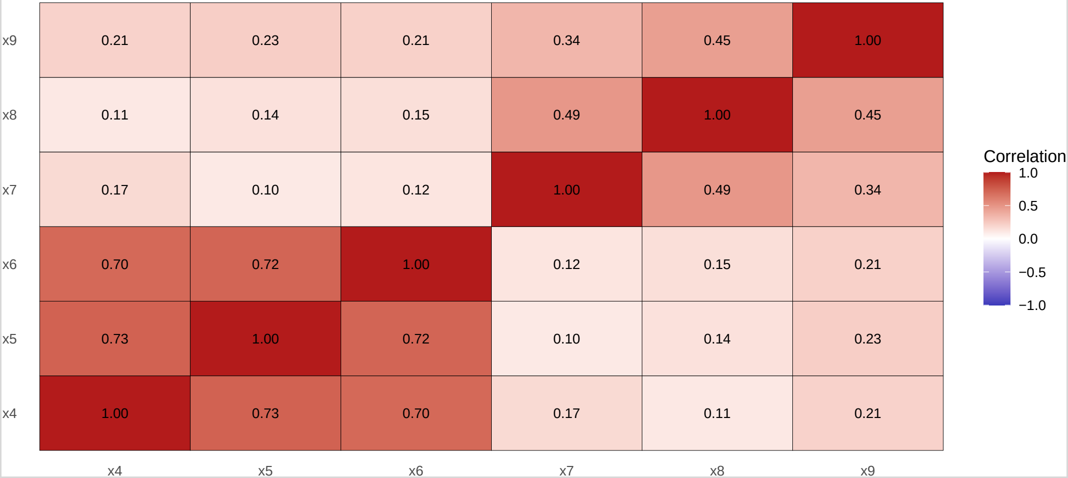
1. **Paragraph (x4)**: Reading comprehension. Students read short paragraphs and answer questions about the main idea, details, and simple inferences.
2. **Sentence (x5)**: Sentence completion. Choose the word or phrase that best completes a sentence so it is clear and grammatical; emphasizes vocabulary-in-context and syntax.
3. **Word Meaning (x6)**: Vocabulary. Select the best synonym/definition for a target word; reflects breadth of lexical knowledge.
4. **Addition (x7)**: Timed arithmetic fluency. Complete many simple addition problems quickly and accurately; captures processing speed with minimal reasoning load.
5. **Counting Dots (x8)**: Rapid visual enumeration. Quickly count small groups of dots across many items; measures visual scanning and speeded accuracy.
6. **Straight–Curved Capitals (x9)**: Perceptual discrimination speed. Scan capital letters and mark those with only straight strokes (versus any curved strokes), under strict time limits.

Assessing Factorability

We start our factor analysis process by assessing the factorability for tests x4–x9. First we collect all the required information.

We now show the correlation heatmap:

Correlation heatmap for tests x4−x9



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As we can see, all tests so far indicate that factor analysis is appropriate for this dataset.

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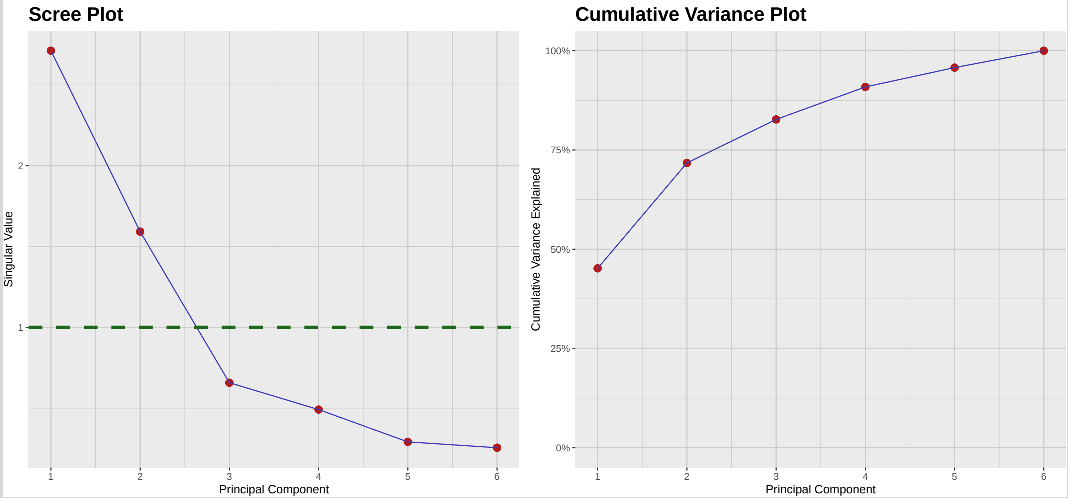
Since we are using the correlation matrix, we will look for singular values greater than 1 (Kaiser criterion) and also look for an “elbow” in the scree plot.

Scree Plot and Cumulative Variance Plot



Notice that both the scree plot and the cumulative variance plot suggest that a two-factor solution is appropriate for analysing the six tests included in this dataset.

Extracting, Rotating, and Scoring Factors

Even though extracting, rotating and scoring factors are theoretically separate steps, in practice they are often performed together using specialized software or statistical packages. Here we will use the `psych::fa` function to perform all three steps at once. We will fit a 2-factor model using maximum likelihood (ML) estimation, compute factor scores using the regression method, and perform a varimax rotation to simplify the factor structure.

Assessing Model Fit

We now evaluate how well the 2-factor ML solution reproduces the observed correlations. For this, we perform the tests χ-squared, RMSEA, CFI, TLI, and RMSR.

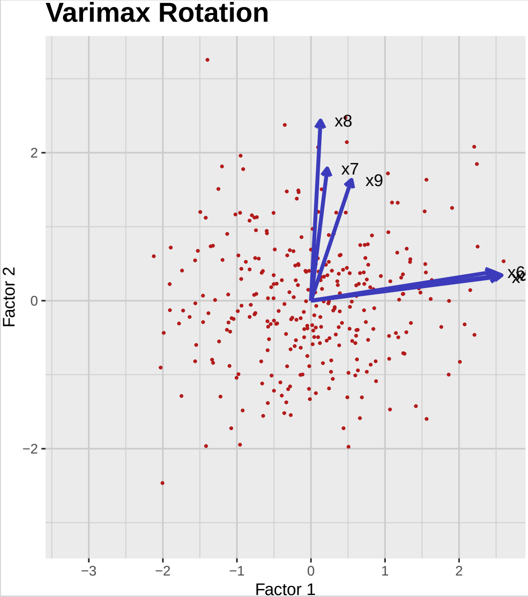
The results of these tests are:

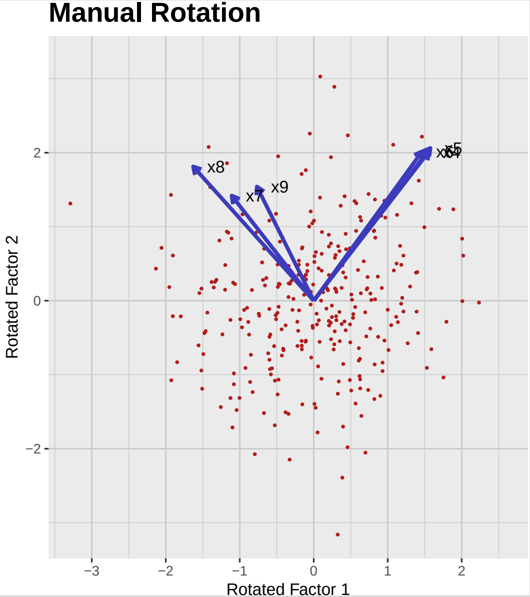
$$\begin{aligned} \chi^2 \text{ Test} &= 0.1156716 \\ \text{RMSEA} &= 0.0531269 \\ \text{TLI} &= 0.9804573 \\ \text{RMSR} &= 0.0411927 \end{aligned}$$

All tests indicate that the 2-factor model fits the data well.

Visualizing the Model

Finally, we will visualize the model using a biplot. We will also rotate the factors manually to illustrate what could happen if we did not choose an appropriate rotation method.


Varimax Rotation
Factor 2
Factor 1
x8
x7
x9
2
0
−2
−3
−2
−1
0
1
2


Manual Rotation
Rotated Factor 2
Rotated Factor 1
x8
x9
x7
2
0
−2
−3
−2
−1
0
1
2